International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE

Final Examination, Autumn 2024

Course Code: CSE 3631 **Course Title: Operating Systems**

Time: 2 Hours 30 Minutes Total Marks: 50

[Answer *all* the following questions. Figures in the right hand margin indicate full marks. Use Separate answer scripts for Group-A and Group-B. Course Outcomes and Bloom's Levels are mentioned in additional Columns]

Bloom's Levels of the Questions						
Letter Symbols	R	U	App	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

CO	CLO Description	PO
1.	**Understand** the basics of operating system.	1
2.	**Analyze** the performance of various CPU scheduling algorithms and solve process synchronization and deadlock problems.	2
3.	**Examine** the efficiency aspect of using system resources (processor, memory, file and disk)	2

CLO BL Marks

Group A

a) Explain the concept of "Deadlock Detection" and how it differs from deadlock prevention and avoidance. Use the wait-for graph method to determine if a deadlock exists. CO1 U 03

OR,

Define and explain the purpose of the Resource-Allocation Graph (RAG) in detecting deadlocks. Given the following RAG, identify if there is a deadlock, and if so, which processes and resources are involved:

- Processes: P1, P2, P3
- Resources: R1, R2

Edges: P1 -> R1, R1 -> P2, P2 -> R2, R2 -> P3, P3 -> R1

b) *In a multi-process system, consider three resource types (A, B, and C) with initial availability of (6, 8, 7) units respectively, and five processes (P1, P2, P3, P4, P5) with different maximum needs and current allocations as shown below. Using the Banker's algorithm, calculate the need matrix, available resources, and determine whether the current state is safe. Demonstrate the allocation process with a clear example* CO2 Ap 04

Process	Allocation	Max
P1	2 1 2	4 2 3
P2	1 2 3	3 4 4
P3	2 1 1	5 3 3
P4	3 1 4	6 3 5
P5	1 2 2	2 3 3

) Consider a system snapshot with five processes (P0 through P4) and three resource types (A, B, and C) with 3 instances of A, 2 instances of B, and 2 instances of C available. The allocation matrix is as follows: P0 has (0, 1, 0), P1 has (2, 0, 0), P2 has (3, 0, 3), P3 has (2, 1, 1), and P4 has (0, 0, 2). The request matrix is as follows: P0 requests (0, 0, 0), P1 requests (2, 0, 2), P2 requests (0, 0, 0), P3 requests (1, 0, 0), and P4 requests (0, 0, 2). The available resource vector is (3, 2, 2). Using the deadlock detection algorithm, determine if a deadlock exists in the system. Begin by listing the initial Work and Finish vectors, then iterate through the processes to update these vectors. Show each step of the detection process and identify any deadlocked processes. Finally, explain how the algorithm works and the significance of the Work and Finish vectors in detecting deadlocks. CO3 An 03

) Consider a swapping system in which memory consists of the following hole sizes in memory order: 10 MB, 4 MB, 20 MB, 18 MB, 7 MB, 9 MB, 12 MB, and 15 MB. Which hole is taken for successive segment requests of 12 MB,10 MB and 9 MB or best fit? Compare the time complexity of best fit, first fit and worst fit. CO3 E 03

Or

Define contiguous memory allocation and explain its main advantages and disadvantages. Provide examples to illustrate external and internal fragmentation.

) Describe the role of the Translation Look aside Buffer (TLB) in a paging system. How does it improve the efficiency of address translation. CO2 An 04

OR

Given the following paging system parameters and page table, calculate the physical address for the logical address 13. n=2 (number of bits for the page offset), m=4 (number of bits for the page number), 32-byte memory (total memory size), 4-byte pages (page size), 8 pages (total number of pages)

Page Number	Frame Number
0	5
1	2
2	1
3	2
4	3
5	4
6	6
7	0

) Consider a system with the following sequence of memory allocations and deallocations: Allocate 10 KB, Allocate 20 KB, Deallocate 10 KB, Allocate 5 KB, Allocate 15 KB. If the system initially has 50 KB of free memory, demonstrate how each memory allocation strategy (first-fit, best-fit, worst-fit) would handle these requests. Discuss the resulting fragmentation for each strategy. CO3 E 03

Group B

) Explain the concept of "virtual memory" and its importance in modern operating systems. How do paging and segmentation contribute to virtual memory management? Discuss the advantages and potential issues associated with each technique. CO2 E 03

OR

Explain how the logical address is mapped to physical address with TLB.

Discuss the concept of "demand paging" in virtual memory systems. How does it differ from pre-paging? Describe the steps involved in handling a page fault and the impact on system performance. CO1 U 03

c) Explain the Optimal Page Replacement algorithm in the context of memory management. Using the provided sequence of page references and a memory capacity of 4 pages, demonstrate how the Optimal algorithm manages pages in memory. Calculate the page fault rate for the given sequence and compare it to the LRU and FIFO algorithms' performance. Discuss the advantages and limitations of the Optimal algorithm in memory management. CO2 An 04

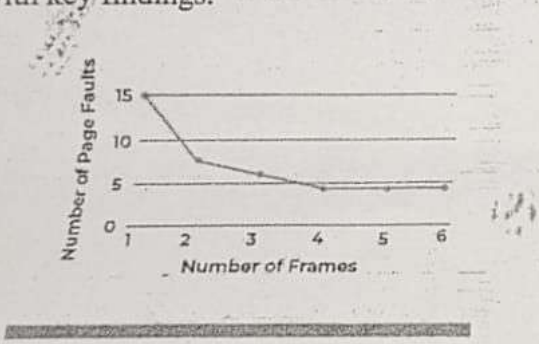
Page Reference String: 4, 2, 3, 1, 2, 4, 5, 1, 2, 3, 4, 5, 2, 1, 4

a) Describe the concept of space allocation methods in file systems. Explain the contiguous, linked, and indexed allocation methods. Provide examples of each and discuss their advantages and limitations in terms of memory utilization and access efficiency. CO2 An 04

b) Explain the concept of acyclic-graph structured directories in file systems and operating systems. How do they differ from traditional hierarchical directory structures, and what advantages do they offer in terms of organizing and accessing files? Provide an example scenario where an acyclic-graph structured directory could be beneficial, and discuss the challenges and considerations in implementing and managing such a directory system. CO2 Ap 03

OR

The following chart represents the number of page fault vs number of frames. Analysis this chart with key findings.



Consider the following scenarios and identify which type of security violation is being described: CO3 E 03

1. A hacker intercepts and reads confidential emails between employees in a company.
2. An attacker modifies the content of a financial transaction to transfer money to their account.
3. A cyber-attack renders a company's website unavailable to users for an extended period.
4. Unauthorized access is gained to a server to use its processing power for mining cryptocurrencies.
5. A large volume of traffic is sent to a server, overwhelming it and causing legitimate users to be unable to access the service.

An employee accidentally shares sensitive customer information with a third-party vendor without proper authorization.

Write short notes on 1) principle of least privilege and 2) protection rings CO3 U 03

)	Explain how denial of service attack act as a network threat ?	CO3	U	02
)	A university computer system has three domains: **Domain 1:** Administrators, **Dómain 2:** Professors ,**Domain 3:** Students There are four objects in the system: **Object 1:** Student Grades File, **Object 2:** Course Materials Directory ,**Object 3:** System Configuration File, **Object 4:** University Website The current access matrix is incomplete. You are tasked to define the access rights for each domain to ensure a secure system while allowing users to perform their tasks. **Access Rights: Read (R),Write (W),Execute (X), None (-)** **Problem:** 1. Define the access rights for each domain (row) to each object (column) in the access matrix. Consider the following: o Administrators should have full control over the system. o Professors need to read course materials and student grades related to their courses. They should not modify student grades or the system configuration. o Students need to read their own grades and access course materials relevant to their enrollment. They should not have access to other students' grades or the system configuration. o The university website should be readable by everyone. ***Or*** Alice wants to send a confidential message (represented as a numerical value, m) to Bob using the RSA encryption algorithm. Given: Two large prime numbers: p = 17 and q = 11 1. Calculate the public and private key 2. Encrypt the message m = last three digits of your ID, using the public key . 3. Decrypt the ciphertext obtained in step 2 using the private key .	CO3	E	05